TOPIC FOUR: MOMENTS

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Contents

0.1 Objectives

By the end of the topic, the learner should be able to:

- Distinguish between the different types of moments.
- Express central moments in terms of moments about an arbitrary point.
- Compute moments for both ungrouped and grouped data.

0.2 Introduction

Moments are popularly used to describe the characteristic of a distribution. The Greek letter μ (read as mu) is generally used to denote moments.

0.3 Raw Moments

Suppose a variable x changes values $x_1, x_2, x_3, \ldots, x_m$ then:

$$\bar{X}^m = \frac{x^n + x_1^n + \ldots + x_n^n}{m} = \bar{X}^m$$

$$\frac{\sum_{i=1}^{n} x_i^n}{m}$$

is called the n^{th} moment about the origin (raw moment).

Example 4.1

$X = 10, 11, 12$

$\bar{X}^2$ is called 2^{nd} moment about the origin:

$$\bar{X}^2 = \frac{10^2 + 11^2 + 12^2}{3}$$

$\bar{X}$ is called the first moment about the origin (mean, $E(X)$):

$$\bar{X} = \frac{10 + 11 + 12}{3}$$

0.4 Central Moments

Given a data set $x_1, x_2, x_3, \dots, x_n$, the r^{th} moment of a variable X about the arithmetic mean is given by:

$$M_r = \frac{\sum_{i=1}^{n}(x_i - \bar{X})^r}{n} \quad \dots\dots\dots\dots\dots\dots\dots\dots \text{(i)} \quad \text{(ungrouped data)}$$

Example 4.2

$X = 10, 11, 12$

First moment about mean:

$$\bar{X} = 11\tfrac{2}{3}$$

$$\frac{\left(10 - 11\frac{2}{3}\right) + \left(12 - 11\frac{2}{3}\right) + \left(13 - 11\frac{2}{3}\right)}{3} = 0$$

N/B: The first moment about the mean is always 0.

Proof: 1^{st} moment about mean is zero

$$M_r = \frac{\sum_{i=1}^{m} x_i - \bar{X}^r}{m}$$

$$= \frac{\sum_{i=1}^{m} x_i}{m} - \frac{\sum_{i=1}^{m} \bar{X}}{m}$$

$$= \bar{X} - \bar{X} = 0$$

Second moment about the mean:

$$M_2 = \frac{\sum_{i=1}^{m}(x_i - \bar{X})^2}{m} = S^2 \quad \text{or variance}$$

0.5 Moments about an Arbitrary Point

The r^{th} moment of a variable X about any arbitrary point A is given by:

$$M'_r = \frac{\sum_{i=1}^{m}(x_i - a)^r}{m} \quad \dots\dots\dots\dots\dots\dots\dots\dots \text{(ii)}$$

Example 4.3

$X = 10, 12, 13,\ a = 10$ (First moment about 10):

$$M'_r = \frac{(10-10)+(12-10)+(13-10)}{3}$$

$$= \frac{2+3}{3} = \frac{5}{3}$$

Remarks

$E(X^r)$ — r^{th} moment about the origin:

$$\frac{\sum x^r}{m}$$

r^{th} Moment about origin:

$$M_1 = 0$$
$$M_2 = S^2$$
$$\vdots$$
$$M_r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^r}{n}$$

Moments about any point a:

$$M'_r = \frac{\sum_{i=1}^{n}(x_i - a)^r}{n}$$

Example 4.4

Find the first, second, third & fourth moments of the following set of numbers about the origin:

$2, 3, 7, 8, 10$

$$M_1 = \bar{X} = \frac{2+3+7+8+10}{5} = 6$$

$$M_2 = \bar{X}^2 = \frac{2^2+3^2+7^2+8^2+10^2}{5} = 45.2$$

$$M_3 = \bar{X}^3 = \frac{2^3+3^3+7^3+8^3+10^3}{5} = 378$$

$$M_4 = \bar{X}^4 = \frac{2^4+3^4+7^4+8^4+10^4}{5} = 3318.8$$

0.6 Relationship between Central Moments and Moments about an Arbitrary Point

Prove that: $M_2 = M_2' - [M_1']^2$

Let $d = x - a$, so $x = a + d$, $\bar{X} = a + \bar{d}$, and $x - \bar{X} = d - \bar{d}$.

$$M_2 = \frac{\sum(x - \bar{X})^2}{n}$$

$$= \frac{\sum(d - \bar{d})^2}{n} = \frac{\sum(d^2 - 2d\bar{d} + \bar{d}^2)}{n}$$

$$= \frac{\sum d^2}{n} - 2\bar{d}\left(\frac{\sum d}{n}\right) + \frac{\sum(\bar{d})^2}{n}$$

$$= \frac{\sum d^2}{n} - 2\bar{d}\left(\frac{\sum d}{n}\right) + (\bar{d})^2$$

$$= \frac{\sum d^2}{n} - 2\bar{d} \cdot \bar{d} + (\bar{d})^2$$

$$= \frac{\sum d^2}{n} - (\bar{d})^2$$

$$= \left(\frac{\sum(x - a)}{n}\right)^2 - \left(\frac{\sum(x - a)}{n}\right)^2$$

$$= M_2' - \left(\frac{\sum d}{n}\right)^2 = M_2' - \left(\frac{\sum(x - a)}{n}\right)^2$$

$$\boxed{M_2 = M_2' - [M_1']^2}$$

Prove that: $M_3 = M_3' - 3M_1'M_2' + 2(M_1')^3$

$$M_3 = \frac{\sum(x - \bar{X})^3}{n} = \frac{\sum(d - \bar{d})^3}{n}$$

$$= \frac{\sum\left(d^3 - 3d^2\bar{d} + 3d(\bar{d})^2 - (\bar{d})^3\right)}{n}$$

$$= \frac{\sum d^3}{n} - 3\bar{d}\left(\frac{\sum d^2}{n}\right) + 3(\bar{d})^2\frac{\sum d}{n} - (\bar{d})^3$$

$$= \frac{\sum d^3}{n} - 3\bar{d}\left(\frac{\sum d^2}{n}\right) + 3(\bar{d})^2\bar{d} - (\bar{d})^3$$

$$= \frac{\sum d^3}{n} - 3\bar{d}\left(\frac{\sum d^2}{n}\right) + 2(\bar{d})^3$$

$$= \bar{d^3} - 3\bar{d} \cdot \bar{d^2} + 2(\bar{d})^3$$

$$= \left(\frac{\sum(x-a)}{n}\right)^3 - 3\left(\frac{\sum(x-a)}{n}\right)\left(\frac{\sum(x-a)^2}{n}\right) + 2\left(\frac{\sum(x-a)}{n}\right)^3$$

$$\boxed{M_3 = M_3' - 3M_1'M_2' + 2(M_1')^3}$$

Prove that: $M_4 = M_4' - 4M_1'M_3' + 6(M_1')^2M_2' - 3(M_1')^4$

$$M_4 = \frac{\sum(x-\bar{x})^4}{n} = \frac{\sum(d-\bar{d})^4}{n}$$

$$= \frac{\sum\left(d^4 - 4d^3(\bar{d}) + 6d^2(\bar{d})^2 - 4d(\bar{d})^3 + (\bar{d})^4\right)}{n}$$

$$= \bar{d^4} - 4\bar{d} \cdot \bar{d^3} + 6(\bar{d})^2 \cdot \bar{d} - 4(\bar{d})^3 \cdot \bar{d} + (\bar{d})^4$$

$$= \bar{d^4} - 4\bar{d} \cdot \bar{d^3} + 6(\bar{d})^2 \cdot \bar{d} - 4(\bar{d})^4 + (\bar{d})^4$$

$$= \bar{d^4} - 4\bar{d} \cdot \bar{d^3} + 6(\bar{d})^2 \cdot \bar{d} - 3(\bar{d})^4$$

$$= \left(\frac{\sum(x-a)}{n}\right)^4 - 4\left(\frac{\sum(x-a)}{n}\right)\left(\frac{\sum(x-a)^3}{n}\right) + 6\left(\frac{\sum(x-a)}{n}\right)^2\left(\frac{\sum(x-a)^2}{n}\right) - 3\left(\frac{\sum(x-a)}{n}\right)^4$$

$$\boxed{M_4 = M_4' - 4M_1'M_3' + 6(M_1')^2M_2' - 3(M_1')^4}$$

For Grouped Data

The r^{th} moment about the origin is given by:

$$\bar{X}^r = \frac{f_1x_1^r + f_2x_2^r + ... + f_nx_n^r}{f_1 + f_2 + ... + f_n} = \frac{\sum_{i=1}^n f_ix_i^r}{\sum f_i}$$

r^{th} **Moment about the mean:**

$$M_r = \frac{\sum_{i=1}^n f_i(x_i - \bar{X})^r}{\sum f_i}$$

For different values of r, we shall get different moments. Thus if we put $r = 1$, we will get the first moment, if we put $r = 2$, we will get the second moment, and so on.

$$M_1 = 0 \qquad M_2 = S^2$$

r^{th} **Moment about any point** a**:**

$$M'_r = \frac{\sum_{i=1}^{n} f_i (x_i - a)^r}{\sum f_i}$$

If $X_i = A + CU_i$ where A and C are constants, then:

$$M'_r = \frac{C^r \sum f_i u_i}{\sum f_i}$$

Example 4.5

The marks obtained by students in a class were as follows:

Mark	Frequency	X	D	U = d/c, c = 10	fu	fu^2	fu^3	fu
1-10	4	5.5	-40	-4	-16	64	-256	1024
11-20	5	15.5	-30	-3	-15	45	-135	405
21-30	32	25.5	-20	-2	-64	128	-256	512
31-40	89	35.5	-10	-1	-89	89	-89	89
41-50	102	45.5	0	0	0	0	0	0
51-60	78	55.5	10	1	78	78	78	78
61-70	63	65.5	20	2	126	252	504	1008
71-80	21	75.5	30	3	63	189	567	1701
81-90	9	85.5	40	4	36	144	576	2304
91-100	3	95.5	50	5	15	75	375	1875
Total	406	134			134	1064	1364	8996

Obtain the first 4 moments about the mean.

Moments about 45.5:

$$M'_1 = \frac{C \sum fu}{\sum f} = \frac{10 \times 134}{406} = 3.30$$

$$M'_2 = \frac{C^2 \sum fu^2}{\sum f} = \frac{100 \times 1064}{406} = 262.07$$

$$M'_3 = \frac{C^3 \sum fu^3}{\sum f} = \frac{1000 \times 1364}{406} = 3359.61$$

$$M'_4 = \frac{C^4 \sum fu^4}{\sum f} = \frac{10000 \times 8996}{406} = 221576.35$$

$$M_1 = 0$$

$$M_2 = M'_2 - [M'_1]^2 = 262.07 - (3.30)^2 = 251.18$$

$$M_3 = M'_3 - 3M'_1 M'_2 + 2(M'_1)^3 = 3359.61 - 3(3.30)(262.07) + 2(3.30)^2 = 786.90$$

$$M_4 = M_4' - 4M_1'M_3' + 6(M_1')^2M_2' - 3(M_1')^4$$
$$= 221576.35 - 4(3.30)(3359.35) + 6(3.3)^2(262.07) - 3(3.30)^4 = 194353.15$$

Example 4.6

Calculate the first four moments about the mean of the following distribution:

x	f	d (x - a)	u	fu	fu²	fu³	fu
12	1	-6	-3	-3	9	-27	81
14	4	-4	-2	-8	16	-32	64
16	6	-2	-1	-6	6	-6	6
18	10	0	0	0	0	0	0
20	7	2	1	7	7	7	7
22	2	4	2	4	8	16	32
Total	30			-6	46	-42	190

$$M_1' = \frac{C\sum fu}{\sum f} = \frac{2\times(-6)}{30} = -0.4$$

$$M_2' = \frac{C^2\sum fu^2}{\sum f} = \frac{4\times 46}{30} = 6.13$$

$$M_3' = \frac{C^3\sum fu^3}{\sum f} = \frac{8\times(-42)}{30} = -11.2$$

$$M_4' = \frac{C^4\sum fu^4}{\sum f} = \frac{16\times 190}{30} = 101.33$$

$$M_1 = 0$$

$$M_2 = M_2' - [M_1']^2 = 6.13 - (-0.4)^2 = 5.97$$

$$M_3 = M_3' - 3M_1'M_2' + 2(M_1')^3 = (-11.2) - 3(-0.4)(6.13) + 2(-0.4)^3 = -3.524$$

$$M_4 = M_4' - 4M_1'M_3' + 6(M_1')^2M_2' - 3(M_1')^4$$
$$= 101.33 - 4(-0.4)(-11.2) + 6(-0.4)^2(6.13) - 3(-0.4)^4 = 89.294$$